(a) Proportional Relationships (StructPF Basketball Task 18)

Student Task (Shortened)

Two shooting drills, each recorded as timed segments (minutes, points).

Drill A: (2, 4), (3, 6), (5, 10)

Drill B: (2, 3), (3, 8), (5, 9)

"Which drill shows that points are proportional to minutes?"

Familiar Grade 6 Method (Overall Rate per Drill):

A: 20 pts / 10 min = 2 B: 20 pts / 10 min = 2 → Tie

Left for Students (Constant Rate in Every Segment):

A: 2, 2, 2 (proportional) B: 1.5, 2.7, 1.8 (not proportional)

PF Eval: Yes on All Five Measures

(b) Percent Discounts (StructPF Shopping Task 26)

Student Task (Shortened)

An $80 jacket; both stores add the same 8% sales tax.

Store A: 20% off, then 10% off the reduced price

Store B: 30% off once

Friend: "20% off and then 10% off is the same as 30% off."

"Which store will give you the lower final price?"

Familiar Shortcut (Add the Percentages):

20% + 10% = 30% → A = B = $60.48 → Tie

Left for Students (Second Discount on the Reduced Price):

A: $80 → $64 → $57.60 → $62.21 B: $80 → $56 → $60.48

PF Eval: Yes on All Five Measures